SAMPLE

Cosmic Explorers: Spacesuits

Personal Spacecraft for the Human Body

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Free sample: Introduction & Chapter 1

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Introduction

A spacesuit is a small spacecraft built in the shape of a person. It holds air against the vacuum, carries its own weather, fends off the sun's raw heat and the cold of the shadow, wraps its passenger in enough shielding to survive the micrometeoroids that cross the void faster than any rifle round, and does all of it while staying soft enough for a human hand to close around a tool. The craft is worn rather than flown, but the definition holds: pressure hull, life support, communications, navigation aids, and a crew of exactly one, packed into a garment that can be folded and hung on a hook.

This book follows that garment from its beginnings in high-altitude balloons and test aircraft to the moonwalks of the Apollo years, the long working suits of the station era, and the new commercial and national suits now being stitched for the return to the Moon and the first attempts on Mars. The story is told as a craft's biography: a lineage of sewers, engineers, and test subjects, mostly uncelebrated, who discovered by trial what a body needs in a place designed to kill it. The account keeps its honesty in the open: the numbers quoted are the ones the programs published, the failures are named as plainly as the successes, and where the field still argues, the argument is shown rather than settled.

One caution belongs at the front door. A suit is never one thing; it is a stack of trades, every gain in one protection bought at a cost in another, the pressure that keeps the blood boiling also stiffening every joint, the shielding that stops the grit also adding the mass that breaks the back. The biography of the spacesuit is the story of a century of bargaining, thread by thread, with a hostile sky.

Chapter 01: The Discovery of Spacesuits

The suit began on Earth, at altitudes where the air itself gave out. The first chapter of the discovery is not the story of space but the story of the stratosphere: the pilots and balloonists of the 1930s who climbed past the height where a lung can pull a breath, and the tailors and tinkerers who tried to bring a survivable atmosphere along with them.

The founding name belongs to an American aviator of the 1930s, a record-setting pilot who lost the sight in one eye to an oil rig accident and flew on anyway, around the world twice. To survive the altitudes his records demanded, he worked with a defense contractor's engineer to build the first practical pressure suit, a rubberized garment blown up like a tire to hold the pilot's body at pressure, clumsy in the extreme, so stiff when inflated that the man could barely work his controls, but it carried him to the high thin air and back, and the principle it proved became the parent of every suit since: if the atmosphere will not come with you, wear it.

The war years matured the craft under other names, the high-altitude bombers and the rocket planes of the next decade demanding partial-pressure garments that squeezed the pilot's body mechanically, and the 1950s saw the specialty firms emerge, the seamstresses of the parachute and brassiere trades discovering that their craft of soft complex structures, built in layers to hold a shape against a pressure, was exactly the craft the rocket planes needed. When the first astronauts were chosen, their suits came from the same lineage: the Mercury fliers wore modified navy high-altitude garments, pressure-tested leftovers of the jet age, and the founders of the Russian program wore a closely related garment of their own on the first human flight of 1961, the suit there serving a second purpose, its orange hue chosen to make a downed pilot findable in the taiga.

The two programs then learned the same lesson in the same decade, each in its own accident: the suit is not decoration. The Soviet loss that killed a returning crew in 1971, a cabin venting its air on the way home with no suits to hold a single breath, changed the rules on the other side of the world forever, every Soyuz crew from that year on wearing a rescue suit for launch and landing, the habit continuing to the present day. The American program learned its own version on the ground, in a capsule fire during a plugs-out test, after which nothing flammable was allowed against a crew's skin again, the suits included. The card files the pair of lessons as the discovery chapter's own founding receipt: the suit earns its weight only when the day goes wrong, and every program that treated it as a costume paid for the lesson.

The chapter's last discovery is the one the whole book rests on: the suit that walks. The first spacewalks of the middle 1960s, one from each program, proved that a human could work outside, and both walks nearly ended the hard way, the first walker finding his suit ballooned into a rigid cylinder by the vacuum, fighting his way back through the hatch with a fraction of the expected air to spare, and the American walker of the same year cooking inside a hard-worked suit with a fogged-over visor, barely completing the simplest task. The two narrow escapes

were the birth announcements of suit engineering as its own discipline, the field this book follows: not the sewing of a garment, but the design of a one-person spacecraft whose engine room is a backpack and whose controls are human joints.

The middle years of the discovery also carried the suit's first data harvest, the test flights of the rocket aircraft. The research planes of the high desert flew above the bulk of the atmosphere in the 1950s and 1960s, their pilots dressed in the partial-pressure garments of the day, sealed cabins doing most of the protecting and the garment standing as the backup, the whole arrangement teaching the trade its next lesson: a suit that must work only in emergencies can be light and loose, and a suit that must work all day must be engineered like a vessel. The distinction, the emergency garment and the working garment, has organized every suit inventory since, the launch-and-entry suits of every program descending from the first kind and every walking suit from the second.

The discovery chapter also files the trade's first international cross-pollination, the same decade's other program traveling the same road with its own makers. The Russian garment tradition grew from the same high-altitude roots, its stratospheric balloonists and its test parachutists wearing pressure garments before the first satellite flew, and its founding suit, the one worn on the first human orbital flight, was built around a different philosophy from its American cousin: where the navy-derived American garment was a snug custom fit, the Soviet line favored a roomier cut that could be donned quickly and worn long, a bias toward robustness over finesse that the two traditions still carry, visible to this day in the walk-in design of the Russian orbital suit and the tailored assembly of the American one.

The discovery record also keeps the trade's early Hollywood decade, and the card gives it its due, because the first pressure suits were built for flights that were half publicity. The stratospheric balloon projects of the 1930s were expeditions and spectacles both, their suits sewn by girdle and diving firms, tested in the open before crowds, the pilots standing in their rubber armor for newsreel cameras before climbing into capsules that were themselves part pressurized garment, and the decade's flights, several of them to the edge of the atmosphere, gave the trade its first rulebook of real data: at what altitude the lungs fail, at what pressure the blood boils, how long a sealed crew member can work at a given task, the numbers the later rocket programs would one day need already gathered by balloonists whose suits are now museum cloth. The card files the decade's lesson as the trade's own founding paradox: the suit began as a stunt garment, and the stunt gathered the science.

The record also files the suit's first war service. The high-altitude bomber fleets of the Second World War dressed their crews in electrically heated garments and oxygen masks, the pressure suit remaining experimental, and the postwar years converted that wartime craft, the heated suits and the mask plumbing and the altitude physiology having been paid for by air forces, into the rocket-plane era's garments, the trade's lineage running through the war rooms as much as the laboratories, the cold-weather flying boot being a direct ancestor of the insulated overshoe that later stood on the Moon.

The discovery chapter closes its ledger with the road not taken, because the trade's history is a story of forks. One lineage, the one this book follows, pursued the sealed pressure garment, air held inside a fabric hull. A rival school of the same era pursued the sealed cabin, the crew riding in shirtsleeves inside a pressurized vessel with no personal armor at all, and for a decade the cabin school looked like the winner: the capsules of both programs flew their crews in comfort, the suits being demoted to cabin wear and backup. The trade answered with the accident record, the sealed cabin being a single point of failure, and the fork resolved itself across the grimmest ledger entries of the age, the programs that had bet everything on the cabin losing crews when cabins failed, the programs that kept the personal garment gaining the rule that survives to this day: the ship is pressurized, and the crew member wears the backup anyway, the suit having won its argument not in the design studio but in the inquiry reports, the card filing the resolution as the trade's founding constitutional settlement.

One more card closes the discovery chapter, the suit's second-hand market, the trade's most unexpected economy. The flown garments are among the most collectible objects on Earth, and the retired prototypes and spares have scattered into private hands over the decades, the trade's surplus becoming the museums' and the collectors' stock, the card filing the market as the discovery record's footnote with teeth: the garment that began as a stunt suit became a flight article and ended as an artifact, each stage of its life being traded and tracked, the suit being the only machine whose whole lineage, stunt to spacecraft to heirloom, was lived inside a single century.

One last card for the discovery ledger, the trade's debt to its failures' record keepers. The suit's history is unusually well documented because the trade institutionalized its own autopsies, every accident followed by an inquiry whose findings were published, every chamber incident logged, the trade having been honest in public about its worst days to a degree few industries have matched, and the card files the debt as the chapter's own closing

credit: the modern suit is the sum of a century of honest paperwork, the safest garment ever sewn being the best-documented one, and the discovery chapter's final lesson being bookkeeping.